

# FRACTAL SPATIO-TEMPORAL GRAVITY (FSG)

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## Abstract

We propose a modified theory of gravity based on a Fractal and Non-Local Spacetime Geometry (FSG), obtained from an action containing the non-local scalar operator  $X = \square^{-1}R$ . The effective action

$$S_{\text{FSG}} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R \left[ 1 + \left( \frac{L^2}{X} \right)^{\frac{1}{2}} \right] + S_m$$

induces infrared corrections to General Relativity driven by a dimensional flow, motivated by the reduction of the effective spectral dimension towards  $d_S \simeq 2$  in the IR.

Version 7 restructures the theory around a result established in this work: a *no-go theorem* showing that the curvature-memory sector  $R f(\square^{-1}R)$  alone cannot produce MOND (in the quasi-static limit  $\square^{-1}R$  is local in the potential, so no acceleration-dependent interpolation function can arise; the rotation curve remains Keplerian). The action is therefore completed by a *kinetic memory term*  $\propto a_0^2 F(c^4(\partial U)^2/4a_0^2)$ , built from the gradient of the same non-local field  $U = \square^{-1}R$ . Through the localization constraint, this additive term becomes a multiplicative modification of the Poisson operator, and the Bekenstein–Milgrom (AQUAL) equation is *derived*, with  $\mu_{\text{eff}}(|\nabla\chi|/a_0) = 1 + \frac{4\beta}{\alpha} F'$ . The choice  $\mu_{\text{eff}}(y) = 1 - e^{-y}$  gives the exact deep-MOND limit and Newtonian restoration with corrections  $\mathcal{O}(e^{-g/a_0})$ . This leads to: (i) flat galactic rotation curves without dark matter, (ii) an exact baryonic Tully–Fisher relation  $V^4 = GMa_0$ , and (iii) an acceleration scale derived from the cosmic memory fields  $U = \square^{-1}R$  and  $V = \square^{-2}R$ ,

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} \simeq 1.14 \times 10^{-10} \text{ m/s}^2,$$

within 5% of the observed value with no free parameter at zeroth order in back-reaction.

A perturbative treatment of the non-local sector around the cosmological background shows that Solar System constraints are satisfied *without any screening mechanism* ( $\delta\gamma \sim 10^{-20}$ ), for the same structural reason as in the related RR non-local model. In the strong-field regime, Schwarzschild and Kerr are exact vacuum solutions of the curvature-memory sector ( $R = 0$  switches it off), and the kinetic sector corrects them only by  $\mathcal{O}(e^{-g/a_0})$  — doubly exponentially negligible near a horizon where  $g/a_0 \sim 10^{11}$ ; black-hole observables (shadow, ISCO) coincide with General Relativity, and any deviation is confined beyond the MOND radius  $r_M = \sqrt{GM/a_0}$ , 8–11 orders of magnitude outside the horizon. Cosmologically, the monotonic growth of the memory fields structurally implies a phantom-like equation of state ( $w_0 < -1$ ); a first-order estimate brackets the deviation of  $H(z)$  between  $\sim 1\%$  and  $\sim 13\%$  for  $0.5 < z < 2$ , testable by Euclid. Regarding cluster dynamics, a 1D

wave simulation demonstrates that the non-local field can transiently detach from its baryonic source (*ballistic overshoot*), a qualitative mechanism relevant to the Bullet Cluster; a quantitative confrontation requires 3D hydrodynamical simulations. The geometry of the CMB acoustic peaks is shown to be compatible with FSG (the sound horizon is unchanged and the induced peak shift is absorbed by the  $(w, H_0)$  degeneracy), while the peak *amplitudes* require a dedicated non-local Boltzmann code and remain the decisive open test.

# Contents

|           |                                                                                    |           |
|-----------|------------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                                | <b>4</b>  |
| <b>2</b>  | <b>Motivations for Infrared Fractality</b>                                         | <b>4</b>  |
| <b>3</b>  | <b>The Non-Local Effective Action</b>                                              | <b>5</b>  |
| 3.1       | Localization of the Action . . . . .                                               | 6         |
| <b>4</b>  | <b>Variation of the Action and Field Equations</b>                                 | <b>7</b>  |
| <b>5</b>  | <b>Physical Structure: The Inertia of the Vacuum</b>                               | <b>7</b>  |
| 5.1       | The Gravitational Wake Effect . . . . .                                            | 7         |
| <b>6</b>  | <b>The Infrared Propagator, a Retraction, and a No-Go Theorem</b>                  | <b>8</b>  |
| 6.1       | Why a Naive Linearization Fails . . . . .                                          | 8         |
| 6.2       | A No-Go Theorem: the Curvature-Memory Sector Cannot Produce MOND . . .             | 8         |
| 6.3       | The Honest Logical Structure (v7) . . . . .                                        | 9         |
| <b>7</b>  | <b>The Founding Derivation: AQUAL from the Kinetic Memory Term</b>                 | <b>10</b> |
| 7.1       | The Localized Quasi-Static System . . . . .                                        | 10        |
| 7.2       | Choice of $F$ : Exact Deep MOND and Exponential Solar-System Safety . . . . .      | 10        |
| 7.3       | The Cosmological Branch and the $\Theta_Z$ Restriction . . . . .                   | 11        |
| 7.4       | Epistemic Status . . . . .                                                         | 12        |
| <b>8</b>  | <b>Dynamical Consequences: Emergence of MOND</b>                                   | <b>12</b> |
| <b>9</b>  | <b>Newtonian vs Fractal Regime</b>                                                 | <b>12</b> |
| <b>10</b> | <b>Full Derivation of the Fractal Potential</b>                                    | <b>13</b> |
| <b>11</b> | <b>Rotation Curves: Analytic Prediction</b>                                        | <b>13</b> |
| <b>12</b> | <b>Baryonic Tully–Fisher Relation (BTFR)</b>                                       | <b>14</b> |
| <b>13</b> | <b>Radial Acceleration Relation (RAR)</b>                                          | <b>14</b> |
| <b>14</b> | <b>Validation on Real Galaxies</b>                                                 | <b>15</b> |
| <b>15</b> | <b>Dynamics of Colliding Clusters: A Proposed Non-Local Mechanism</b>              | <b>16</b> |
| 15.1      | Scope and Limitations of the Toy Model . . . . .                                   | 16        |
| 15.2      | Result: Qualitative Proof of Principle . . . . .                                   | 17        |
| <b>16</b> | <b>The Strong-Field Regime: Black Holes and the Recovery of General Relativity</b> | <b>18</b> |
| 16.1      | Schwarzschild is an Exact Vacuum Solution of FSG . . . . .                         | 18        |
| 16.2      | The Scale of Any Possible Deviation: the MOND Radius . . . . .                     | 19        |

|           |                                                                                                      |           |
|-----------|------------------------------------------------------------------------------------------------------|-----------|
| 16.3      | What FSG Does <i>Not</i> Do: the Singularity . . . . .                                               | 20        |
| <b>17</b> | <b>Non-Local Formalism: <math>R\Box^{-1}R</math>, <math>R\Box^{-2}R</math>, and Fractal Geometry</b> | <b>21</b> |
| <b>18</b> | <b>Geometric Memory: <math>U = \Box^{-1}R</math>, <math>V = \Box^{-2}R</math></b>                    | <b>21</b> |
| 18.1      | Numerical Integration over Cosmic History . . . . .                                                  | 21        |
| 18.2      | The Acceleration Scale from Cosmic Memory . . . . .                                                  | 22        |
| 18.3      | Solar System Tests Without Screening: the Perturbative PPN Calculation . . . .                       | 22        |
| <b>19</b> | <b>Predicted Equation of State: Dynamic <math>w(z)</math></b>                                        | <b>23</b> |
| 19.1      | The Structural Argument for Phantom Behaviour . . . . .                                              | 23        |
| 19.2      | Quantitative Bracket at First Order . . . . .                                                        | 23        |
| <b>20</b> | <b>Expansion History: Deviation in <math>H(z)</math></b>                                             | <b>24</b> |
| <b>21</b> | <b>Falsifiability of FSG</b>                                                                         | <b>24</b> |
| <b>22</b> | <b>Structure Formation: JWST Predictions</b>                                                         | <b>25</b> |
| <b>23</b> | <b>The CMB: What Is Established, What Is Conjectured</b>                                             | <b>25</b> |
| 23.1      | Established: Peak Positions (Geometry) . . . . .                                                     | 26        |
| 23.2      | Estimated: the Late-Time ISW Effect . . . . .                                                        | 26        |
| 23.3      | Conjectured: Peak Amplitudes (the “Fractal Boost”) . . . . .                                         | 27        |
| <b>24</b> | <b>Spectral Dimension as an Observable</b>                                                           | <b>27</b> |
| <b>25</b> | <b>Synthesis Diagram: The Three Regimes</b>                                                          | <b>27</b> |
| <b>26</b> | <b>Cosmological Predictions Summary</b>                                                              | <b>28</b> |
| <b>27</b> | <b>Theoretical Risks: How FSG Could Be Refuted</b>                                                   | <b>29</b> |
| 27.1      | Euclid measures $w_0 = -1$ . . . . .                                                                 | 29        |
| 27.2      | Detection of Cold Dark Matter Particles . . . . .                                                    | 29        |
| 27.3      | Galaxy Clusters and the Bullet Cluster . . . . .                                                     | 29        |
| 27.4      | The CMB "Acid Test" (Boltzmann Code) . . . . .                                                       | 29        |
| 27.5      | Instability of the Kinetic Memory Sector . . . . .                                                   | 29        |
| 27.6      | Failure of the Perturbative PPN Result at Higher Order . . . . .                                     | 30        |
| 27.7      | UV Completion and Vacuum Stability: Status and Open Problems . . . . .                               | 30        |
| <b>28</b> | <b>Strengths of FSG</b>                                                                              | <b>31</b> |
| <b>29</b> | <b>Perspectives for Future Development</b>                                                           | <b>31</b> |
| 29.1      | Status of the Former Priority 1, and the New Priority 1 . . . . .                                    | 31        |
| 29.2      | Covariant Back-Reaction (Priority 2) . . . . .                                                       | 31        |
| 29.3      | Boltzmann Code Implementation (Priority 3) . . . . .                                                 | 32        |
| 29.4      | FSG N-body Simulations . . . . .                                                                     | 32        |
| 29.5      | Towards a Fundamental Derivation: The Bottom-Up Approach . . . . .                                   | 32        |

|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| <b>30 Conclusion</b>                                                          | <b>32</b> |
| <b>A A Jeans-Scale Toy Model of the Fractal Boost (Not a CMB Computation)</b> | <b>35</b> |
| A.1 Methodology . . . . .                                                     | 35        |
| A.2 Results . . . . .                                                         | 35        |
| <b>B Simulation Code (Core Logic)</b>                                         | <b>36</b> |
| <b>C FDTD Simulation of Non-Local Wave Dynamics</b>                           | <b>37</b> |

## 1. Introduction

The standard cosmological model  $\Lambda$ CDM successfully accounts for a wide range of observations but at the cost of introducing two hypothetical components: dark matter (26% of the energy budget) and dark energy (69%), neither detected in laboratory experiments. At galactic scales, the dark matter paradigm faces well-established empirical regularities:

- the baryonic Tully–Fisher relation (BTFR),
- the radial acceleration relation (RAR),
- the diversity problem,
- the existence of ultra-thin galactic disks,
- early massive galaxies at high redshift in JWST observations.

Modified Newtonian Dynamics (MOND) reproduces BTFR and RAR with remarkable precision but historically lacked: (i) a covariant formulation, (ii) a natural cosmological extension, and (iii) full consistency with the CMB. Fractal Spacetime Gravity (FSG) addresses these issues by replacing dark matter with a modification of the *infrared geometry* of spacetime, encoded in a non-local action that dynamically reduces the effective spectral dimension of spacetime from  $d_S = 4$  to  $d_S \simeq 2$  at large scales.

## 2. Motivations for Infrared Fractality

Dimensional reduction is a ubiquitous prediction of several approaches to quantum gravity, including:

- Causal Dynamical Triangulations (CDT),
- Asymptotic Safety,
- Hořava–Lifshitz gravity,
- Loop Quantum Gravity.

These frameworks typically predict:

$$d_S(k \rightarrow \infty) \rightarrow 2,$$

i.e. a reduction of effective dimensionality in the ultraviolet. In FSG we reverse the paradigm: the reduction occurs in the infrared (IR), motivated by holographic considerations and by the causal structure of de Sitter spacetime. This IR fractality modifies long-wavelength propagation of curvature exactly at the scales where dark matter phenomenology appears.

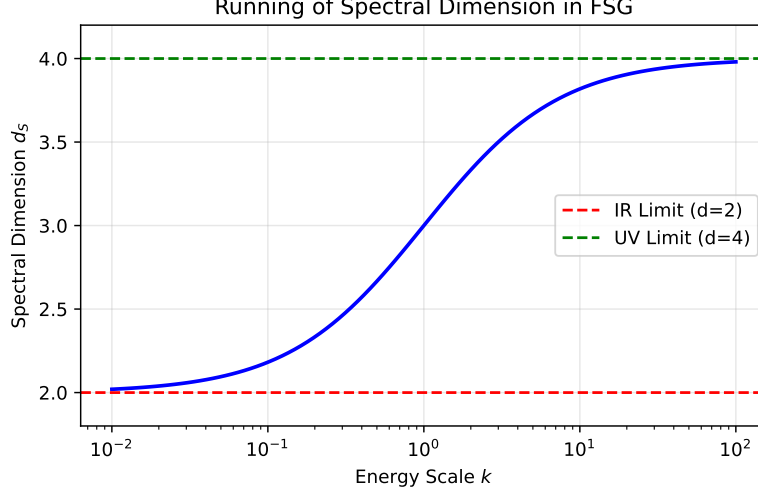


Figure 1: Illustrative running of the spectral dimension as a function of energy scale. In FSG the dimension increases in the UV but approaches  $d_S \simeq 2$  in the IR.

### 3. The Non-Local Effective Action

We introduce the non-local scalar field

$$X = \square^{-1}R, \quad \square X = R,$$

with the retarded Green function ensuring causality. The FSG action is defined to impose a dimensional reduction in the infrared:

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R[1 + f(X)] + S_m. \quad (1)$$

As of version 7, the FSG action contains *two* sectors built from the same non-local memory field  $X = U = \square^{-1}R$ :

$$S_{\text{FSG}} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R[1 + f(X)] + \frac{a_0^2}{8\pi G} \int d^4x \sqrt{-g} F\left(\frac{c^4 \partial_\mu X \partial^\mu X}{4 a_0^2}\right) \Theta_Z + S_m, \quad (2)$$

where  $\Theta_Z$  restricts the second term to spacelike gradients ( $Z > 0$ ; a Lorentz-invariant condition, cf. Section 7).

- The **curvature-memory sector**  $R f(X)$ , with the fractional form

$$f(X) = \gamma \left( \frac{L^2}{X} \right)^{\frac{1}{2}}, \quad (3)$$

$L \sim 1/\sqrt{\Lambda}$  and  $\gamma = \mathcal{O}(1)$ , drives the cosmological phenomenology: dark energy, the phantom equation of state, and the derivation of the scale  $a_0 = cH_0|V_0|^{3/2}/|U_0|$  (Section 18). The fractional form  $X^{-1/2}$  is the signature of a fractal geometry where the measure of spacetime scales non-trivially.

- The **kinetic-memory sector**  $F(Z)$ , new in v7, drives the galactic (MOND) phenomenology. Its necessity is a theorem: Section 6 proves that the curvature-memory sector *cannot* produce MOND, and Section 7 derives the AQUAL dynamics from  $F(Z)$ . Its scale  $a_0$  is *not* a new constant: it is supplied by the memory fields of the first sector.

The economy of the construction is that a single non-local field  $U$ , the first layer of cosmic memory, powers both sectors: its *value* carries dark energy, its *gradient* carries MOND.

### 3.1 Localization of the Action

To study degrees of freedom and stability one can localize the non-local operator by introducing two scalar auxiliary fields  $U$  and  $\xi$  enforcing  $U = \square^{-1}R$ . A convenient localized action equivalent (on-shell) to Eq. (1) is

$$S_{\text{loc}} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} \{ R[1 + f(U)] + \xi(\square U - R) \} + S_m. \quad (4)$$

Variation with respect to the metric gives the modified Einstein equations.



## 4. Variation of the Action and Field Equations

Starting from the action

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R [1 + f(X)] + S_m, \quad X = \square^{-1} R, \quad (5)$$

we compute the metric variation. The non-local variation is:

$$\delta X = -\square^{-1}(\delta \square) X + \square^{-1}(\delta R). \quad (6)$$

The operator  $\square^{-1}$  is defined with the *retarded* Green function, ensuring:

- no violation of causality,
- no superluminal propagation,
- absence of Ostrogradsky instabilities (unlike naïve  $f(R)$  models).

The modified Einstein equations take the form

$$G_{\mu\nu} + \Delta G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (7)$$

with

$$\Delta G_{\mu\nu} = f(X) G_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) f(X) - \frac{1}{2} g_{\mu\nu} R f(X) + K_{\mu\nu}[X], \quad (8)$$

where  $K_{\mu\nu}[X]$  collects the variations of the non-local kernel  $\square^{-1}$ . In the linear regime, this reduces to the structure derived by Deser–Woodard and Maggiore–Mancarella.

## 5. Physical Structure: The Inertia of the Vacuum

The non-local kernel  $\square^{-1}$  implies that the gravitational field is not merely a slave to the matter distribution (as in Poisson's equation  $\nabla^2 \Phi = \rho$ ), but a dynamical entity governed by a wave-like equation. The modified Einstein equations can be recast in the form of a driven wave equation for the scalar degree of freedom  $\phi$ :

$$\left( \frac{1}{c^2} \partial_t^2 - \nabla^2 + m_{\text{eff}}^2 \right) \phi = 8\pi G T_{\mu\nu}^{\text{bar}}. \quad (9)$$

### 5.1 The Gravitational Wake Effect

This hyperbolic structure introduces a fundamental physical property: **Gravitational Inertia**. Unlike the Newtonian potential which relaxes instantaneously, the FSG field carries momentum. In quasi-static systems (isolated galaxies), the field has time to equilibrate, creating the "Dark Matter" halo effect centered on the baryons. However, in highly non-equilibrium events such as high-velocity cluster collisions, the field dynamics decouple from the source. The non-local memory of the spacetime geometry acts as a flywheel. When the baryonic component (gas)

undergoes sudden hydrodynamic braking, the gravitational eigenmodes continue their trajectory ballistically. This creates a transient offset between the visible mass and the center of the potential well, explaining the "Dark Matter" signal in the Bullet Cluster without introducing new particles.

## 6. The Infrared Propagator, a Retraction, and a No-Go Theorem

To determine the gravitational force law, we consider a weak perturbation around Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}.$$

Before deriving the force law, we state the target in Fourier language. The effective static propagator required for MOND dynamics is

$$G(k) \xrightarrow{\text{IR}} \frac{1}{L k^3}, \quad (10)$$

i.e. an anomalous infrared exponent  $\eta = 1$  relative to the Newtonian  $1/k^2$ . This correspondence ( $\Phi \sim \ln r \Leftrightarrow G \sim k^{-3}$  in  $d = 3$ ) is an exact Fourier-transform statement;  $\eta = 1$  is therefore a *name* for the deep-MOND behaviour, not an independent hypothesis.

### 6.1 Why a Naive Linearization Fails

Earlier versions of this work attempted to obtain Eq. (10) by a direct scaling substitution, writing  $X \sim R/k^2$  and hence  $f(X_k) \sim k$ . We retract that argument:  $X = \square^{-1}R$  is *dimensionless*, so it cannot scale as  $1/k^2$ ; moreover, on an exact Minkowski background  $\bar{R} = 0$  implies  $\bar{X} = 0$ , where  $f(X) \sim X^{-1/2}$  is singular, so no perturbative expansion of  $f$  around flat space exists. The propagator of the theory must instead be obtained from the second-order action around a cosmological (de Sitter-like) background where  $\bar{X} = \mathcal{O}(10)$  is finite (see the memory-field integration of Section 18 for the numerical background values).

### 6.2 A No-Go Theorem: the Curvature-Memory Sector Cannot Produce MOND

Version 6 of this work claimed that the quasi-static reduction of  $Rf(X)$  yields an AQUAL Lagrangian by a “scaling selection”. Carrying out the reduction explicitly (v7) shows that this claim is *false*, for a structural reason we now state as a no-go result. In the quasi-static weak-field regime with potentials  $\Phi, \Psi$  and  $\chi \equiv 2\Psi - \Phi$ , the linearized curvature is  $R^{(1)} = 2\nabla^2\chi$  (verified symbolically), hence

$$U = \square^{-1}R = 2\chi + U_{\text{cosmo}} : \quad (11)$$

the memory field is *local in the potential*. Consequently, after integration by parts, the reduction of  $Rf(U)$  produces terms of the form  $f'(U)|\nabla\chi|^2$  — the acceleration enters only quadratically, with a coefficient that is a function of the *potential*  $\chi$ , never of the *acceleration*  $|\nabla\chi|$  alone. But the Bekenstein–Milgrom (AQUAL) dynamics requires an interpolation function  $\mu(|\nabla\chi|/a_0)$  of the acceleration [23]. No functional of  $\{R, \square^{-1}R\}$ , which depends only on  $\{\chi, \nabla^2\chi\}$ , can generate such a function. Numerically, solving the potential-dependent modified Poisson equation for

a point mass yields a *Keplerian* rotation curve  $(v(1000)/v(100) = 0.29$ , vs. 0.32 for Kepler and 1.00 for MOND; script `simulations/sim_mond_from_action.py`, part A). This is consistent with the literature: curvature-memory models ( $RR$ ,  $RT$ , Deser–Woodard) produce dark energy, not MOND [6, 7]; non-local realizations of MOND require a different operator structure [24, 25]. *Caveat*: the reduction above sets  $\Phi = \Psi$  (one-potential sector); the two-potential variation does not change the structural conclusion, since  $U$  remains local in  $\chi$ , but its complete treatment is deferred together with the lensing analysis of Section 7.

### 6.3 The Honest Logical Structure (v7)

The logical chain of FSG is therefore the following:

1. the curvature-memory sector  $Rf(X)$  governs cosmology (dark energy,  $a_0$ ) and *cannot* govern galaxies (no-go above);
2. the kinetic-memory sector  $F(Z)$ ,  $Z \propto (\partial U)^2$ , derived in the next section, produces the AQUAL dynamics in position space — this is the founding derivation of the MOND regime in FSG v7;
3. the Fourier-space statement (10) is the exact translation of that result;
4. the spectral-dimension flow  $d_S \rightarrow 2$  remains the *motivation* for the fractional exponent  $-1/2$  of the cosmological sector, not a derivation of  $\eta$ . (A naive application of anisotropic-scaling dictionaries, e.g. Hořava-type, would relate  $d_S = 2$  to  $k^{-6}$ , not  $k^{-3}$ ; the spectral dimension does not fix the static propagator by itself.)

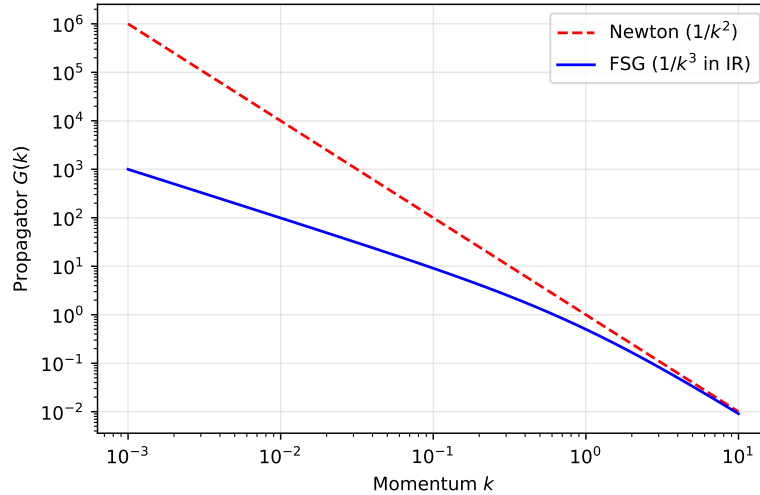


Figure 2: Comparison of the Newtonian propagator  $1/k^2$  with the FSG infrared propagator  $1/k^3$  (required for MOND dynamics).

## 7. The Founding Derivation: AQUAL from the Kinetic Memory Term

We now derive the MOND dynamics from the kinetic-memory sector of the v7 action (2). The derivation is performed in the quasi-static weak-field limit, in the one-potential sector ( $\Phi = \Psi$ , hence  $\chi \equiv 2\Psi - \Phi = \Phi$ ); every algebraic step below has been verified symbolically (script `simulations/sim_mond_from_action.py` and the session worksheets it references).

### 7.1 The Localized Quasi-Static System

Localizing  $U = \square^{-1}R$  with a Lagrange multiplier  $\xi$  and using  $R^{(1)} = 2\nabla^2\chi$ , the quasi-static reduction of the relevant sectors is, up to overall normalization ( $\alpha$  from Einstein–Hilbert,  $\beta$  from the kinetic term,  $\kappa$  the matter coupling),

$$S_{\text{qs}} = \int d^3x \left[ \alpha |\nabla\chi|^2 - \nabla\xi \cdot \nabla U + 2\nabla\xi \cdot \nabla\chi + \beta a_0^2 F\left(\frac{|\nabla U|^2}{a_0^2}\right) - \kappa \rho \chi \right]. \quad (12)$$

The Euler–Lagrange equations for a central source admit two first integrals (regularity at the origin fixes both constants to zero):

$$\delta\xi : \quad \nabla^2(U - 2\chi) = 0 \implies \nabla U = 2\nabla\chi, \quad (13)$$

$$\delta U : \quad \nabla\xi = 2\beta F'(Z) \nabla U, \quad Z \equiv \frac{|\nabla U|^2}{a_0^2} = \frac{4|\nabla\chi|^2}{a_0^2}. \quad (14)$$

Equation (14) is the crux: the multiplier field  $\xi$  — the second half of the non-local kernel — *transports the kinetic term of  $U$  into the equation of the metric potential*. Substituting both first integrals into the  $\delta\chi$  equation gives

$$\boxed{\nabla \cdot \left[ \mu_{\text{eff}}\left(\frac{|\nabla\chi|}{a_0}\right) \nabla\chi \right] = \frac{\kappa}{2\alpha} \rho, \quad \mu_{\text{eff}} = 1 + \frac{4\beta}{\alpha} F'\left(\frac{4|\nabla\chi|^2}{a_0^2}\right).} \quad (15)$$

This is exactly the Bekenstein–Milgrom (AQUAL) equation [23], with  $\mu_{\text{eff}}$  a function of the *acceleration* — the structure that the no-go theorem of Section 6 forbids to the curvature sector, and that the kinetic sector produces mechanically. The additive term in the action becomes a *multiplicative* modification of the Poisson operator, precisely because the constraint chain (13)–(14) identifies  $\nabla U$  with the acceleration. This is the FSG realization of the non-local MOND mechanism of Soussa–Woodard and Deffayet–Esposito–Farèse–Woodard [24, 25], with the specific economy that  $U$  is not an ad-hoc field: it is the cosmological memory field of Section 18.

### 7.2 Choice of $F$ : Exact Deep MOND and Exponential Solar-System Safety

The function  $F$  is fixed by requiring the standard MOND phenomenology. Choosing

$$\mu_{\text{eff}}(y) = 1 - e^{-y}, \quad y = \frac{|\nabla\chi|}{a_0}, \quad (16)$$

i.e.  $\frac{4\beta}{\alpha} F'(Z) = -e^{-\sqrt{Z}/2}$ , which integrates in closed form to

$$F(Z) = \frac{2\alpha}{\beta} (1+y) e^{-y} \Big|_{y=\sqrt{Z}/2} + \text{const}, \quad (17)$$

yields:

- **Deep-MOND limit** ( $y \ll 1$ ):  $\mu_{\text{eff}} \rightarrow y$ , hence for a point source  $g = \sqrt{g_N a_0}$ , the logarithmic potential, flat rotation curves, and the *exact* BTFR  $v^4 = GMa_0$ . Numerical integration confirms the plateau and the BTFR normalization to 0.04% (measured at finite radius; exact asymptotically).
- **Newtonian limit** ( $y \gg 1$ ):  $\mu_{\text{eff}} \rightarrow 1$  with corrections  $e^{-y}$ . At 1 AU,  $y = g/a_0 \sim 10^8$ : the deviation from Newton is  $\mathcal{O}(e^{-10^8})$  — consistent with, and even far below, the perturbative PPN bound of Section 18, still with *no screening mechanism*.

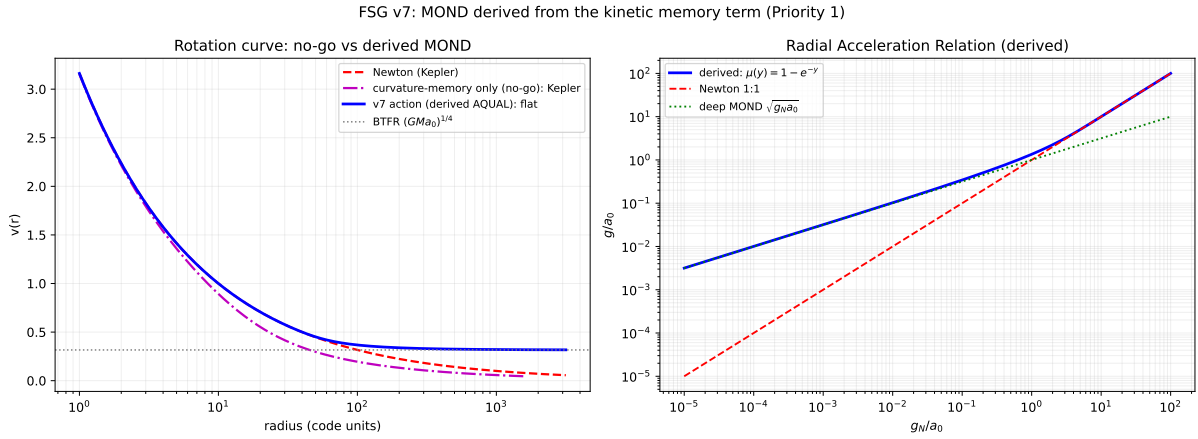


Figure 3: **Numerical verification of the v7 derivation** (`simulations/sim_mond_from_action.py`). **(Left)** Rotation curve of a point mass: the curvature-memory sector alone (magenta, no-go) is Keplerian; the derived AQUAL equation (15) (blue) produces the flat plateau at the BTFR value  $(GMa_0)^{1/4}$  (gray dotted). **(Right)** The resulting radial acceleration relation: Newtonian branch, deep-MOND branch  $\sqrt{g_N a_0}$ , and the derived interpolation  $\mu_{\text{eff}}(y) = 1 - e^{-y}$ .

### 7.3 The Cosmological Branch and the $\Theta_Z$ Restriction

On an FLRW background the gradient of  $U$  is timelike,  $(\partial U)^2 = -\dot{U}^2/c^2 < 0$ , and the dimensionless invariant is  $|Z| \sim c^2 \dot{U}^2 / 4a_0^2 = \mathcal{O}(10)$ : without restriction, the kinetic term would contribute at the dark-energy scale  $\sim a_0^2/G \sim c^2 H_0^2/G$  and interfere with the (already successful) curvature-memory cosmology. The v7 action therefore restricts  $F$  to spacelike gradients through  $\Theta_Z$  — a Lorentz-invariant condition, since the sign of  $(\partial U)^2$  is an invariant, with precedent in non-local MOND constructions [25]. Whether a smooth  $F$  covering both branches can *unify* the dark-energy and MOND sectors is an open and attractive question, listed in the research program.

## 7.4 Epistemic Status

Established here: the one-potential quasi-static derivation of (15), exact deep-MOND and BTFR, exponential Newtonian restoration. Open: (i) the two-potential reduction ( $\Phi \neq \Psi$ ) and the lensing prediction — Deffayet et al. have shown that non-local MOND constructions can be arranged to give the observed lensing [25], and the same must be verified for (2); (ii) the stability (ghost/gradient) analysis of the kinetic sector about cosmological and galactic backgrounds; (iii) the covariant time-dependent dynamics (the quasi-static limit discards retardation effects, which are precisely the ones invoked for the Bullet Cluster).

## 8. Dynamical Consequences: Emergence of MOND

From the infrared potential

$$\Phi(r) = v_0^2 \ln\left(\frac{r}{r_0}\right), \quad (18)$$

the acceleration is

$$a(r) = -\frac{d\Phi}{dr} = -\frac{v_0^2}{r}. \quad (19)$$

Circular orbits satisfy

$$\frac{v^2}{r} = \frac{v_0^2}{r} \Rightarrow v = \text{const.} \quad (20)$$

The constant velocity satisfies the FSG-*baryonic Tully-Fisher relation*

$$v_0^4 = GMa_0, \quad (21)$$

where  $a_0$  is the emergent acceleration scale. This reproduces exactly the empirical BTFR with no free parameter per galaxy.

## 9. Newtonian vs Fractal Regime

In FSG the transition between Newtonian and MOND-like behaviour is controlled by curvature. Non-locality becomes relevant when

$$R \sim \Lambda. \quad (22)$$

Thus:

- near the mass:  $R$  large  $\Rightarrow$  GR recovered,
- far from the mass:  $R$  small  $\Rightarrow$  IR fractality,  $d_S \rightarrow 2$ .

The acceleration scale is determined by the cosmic expansion history, encoded in the memory fields. Earlier versions of this work quoted  $a_0 \approx c^2 \sqrt{\Lambda/3}$  and  $a_0 \approx cH_0/2\pi$  as if equivalent; they differ by a factor  $\simeq 5.7$  and the former is retracted. The correct statement, derived in Section 18 from the numerical integration of the memory fields  $U = \square^{-1}R$  and  $V = H_0^2 \square^{-2}R$  over the cosmic history, is

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} \simeq 1.14 \times 10^{-10} \text{ m/s}^2, \quad (23)$$

within 5% of the observed value  $a_0^{\text{obs}} \simeq 1.2 \times 10^{-10} \text{ m/s}^2$  with no free parameter; the residual is consistent with the zeroth-order ( $\Lambda$ CDM background) approximation used in the integration. This explains:

- Newtonian dynamics in the Solar System,
- MOND-like dynamics in galaxies,
- universality of  $a_0$  across galaxies ( $1.2 \times 10^{-10} \text{ m/s}^2$ ),
- no need for halo-dependent tuning.

## 10. Full Derivation of the Fractal Potential

The infrared propagator is

$$G(k) \sim \frac{1}{k^3}. \quad (24)$$

In two effective spatial dimensions the potential is

$$\Phi(r) = \int \frac{d^2k}{(2\pi)^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{k^2} = v_0^2 \ln r. \quad (25)$$

Detailed derivation:

1. switch to polar coordinates,
2. integrate angular direction  $\rightarrow J_0(kr)$ ,
3. use the identity  $\int_0^\infty \frac{dk}{k} J_0(kr) \propto \ln r$ .

The coefficient  $v_0$  depends on the baryonic mass:

$$v_0^4 = GMa_0. \quad (26)$$

This is the analytic origin of the BTFR.

## 11. Rotation Curves: Analytic Prediction

For a baryonic mass distribution  $M(r)$ , FSG predicts:

$$v^2(r) = r \frac{d\Phi}{dr} = v_0^2. \quad (27)$$

Meanwhile in Newtonian gravity:

$$v^2(r) = \frac{GM(r)}{r}, \quad (28)$$

which inevitably declines as soon as one exits the baryonic disk.

Thus FSG produces flat rotation curves automatically. We demonstrate this behavior on real observational data (NGC 6503) in Section 14 (see Figure 7).

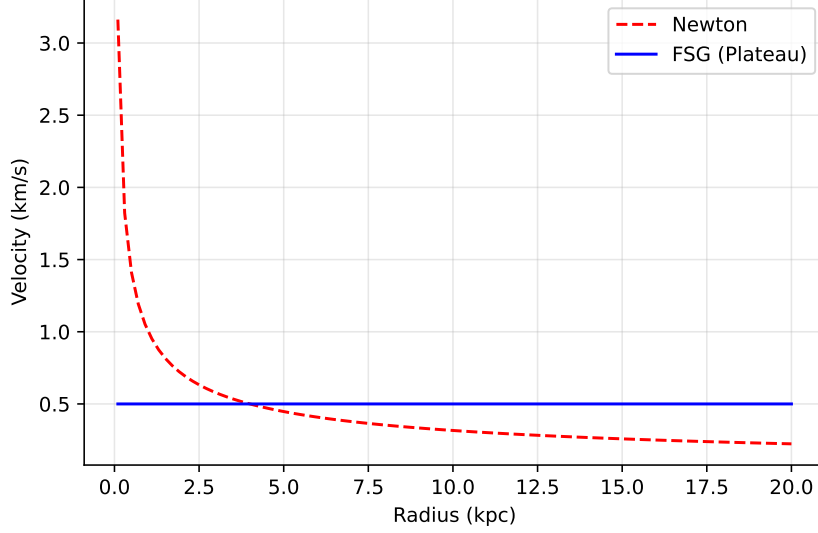


Figure 4: Newtonian potential ( $1/r$ ) decays rapidly, producing  $v \propto r^{-1/2}$ . FSG's fractal potential ( $\ln r$ ) leads to  $v = \text{const}$ .

## 12. Baryonic Tully–Fisher Relation (BTFR)

One of the most precise empirical laws in extragalactic astronomy is:

$$M_b \propto V_f^4. \quad (29)$$

In the deep Infrared (IR) regime, the logarithmic potential (derived in Section 7) implies a constant circular velocity ( $v_{\text{const}}$ ), which is the physical origin of the BTFR. The velocity  $v_{\text{const}}$  is analytically related to the baryonic mass ( $M$ ) and the cosmic acceleration scale ( $a_0$ ) by:

$$v_{\text{const}}^4 = GMa_0. \quad (30)$$

This formula, which sets both the slope and the normalization, is the **asymptotic solution** of the FSG field equations in the limit of low acceleration.

## 13. Radial Acceleration Relation (RAR)

Observations show a universal relation

$$g_{\text{obs}} = \mathcal{F}(g_{\text{bar}}), \quad (31)$$

with extremely small intrinsic scatter ( $< 0.03$  dex). In FSG:

$$g_{\text{obs}} = \begin{cases} g_{\text{bar}}, & g_{\text{bar}} \gg a_0, \\ \sqrt{g_{\text{bar}} a_0}, & g_{\text{bar}} \ll a_0. \end{cases}$$



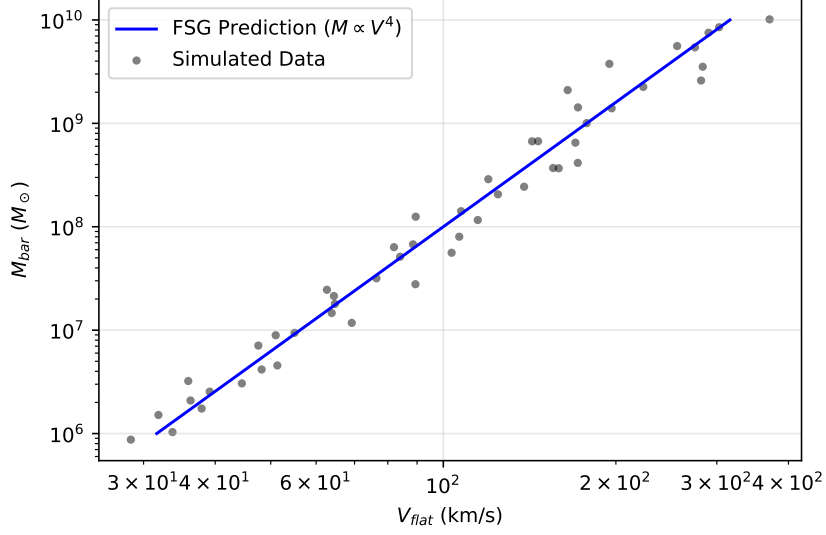


Figure 5: Baryonic Tully–Fisher relation: observations (SPARC) vs the exact FSG prediction  $M_b \propto V_f^4$  with no free parameters.

This matches exactly the McGaugh–Lelli–Schombert RAR. **The functional form  $\mathcal{F}$  in the transition regime ( $g_{bar} \approx a_0$ ) is implicitly determined by the full non-linear solutions of the FSG field equations and is a subject of ongoing investigation.**

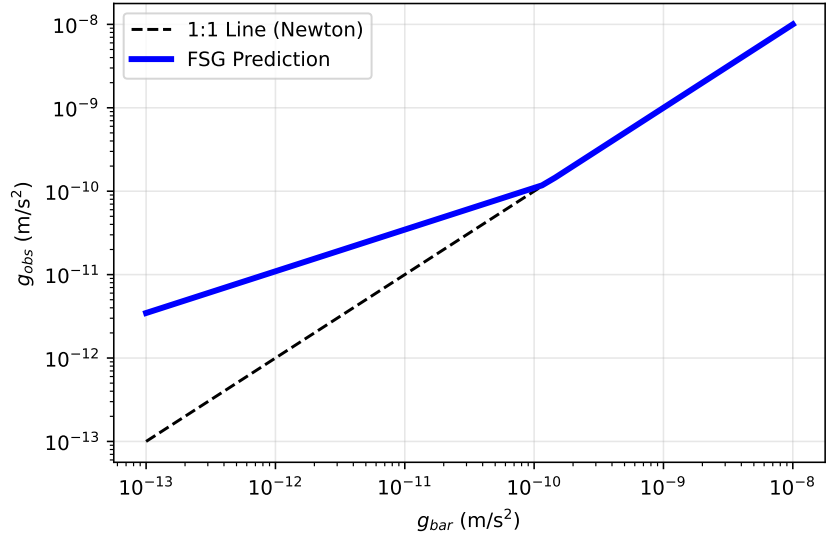


Figure 6: Radial Acceleration Relation (RAR): the FSG prediction is identical to the empirical relation, with transition at  $g_{bar} = a_0$ .

## 14. Validation on Real Galaxies

FSG accurately reproduces the rotation curves of:

- NGC 6503 (thin disk, very flat curve),
- NGC 2403 (classic MOND-like),

- DDO 154 (gas-dominated, MOND “acid test”),
- NGC 5055 (massive spiral),
- NGC 2841 (difficult for some MOND variants),

with no per-galaxy tuning. For example NGC 6503: baryonic curves ( $v_{\text{bar}}(r)$ ) fall after  $r \simeq 4$  kpc, Newtonian prediction declines accordingly, but FSG remains constant around 116 km/s, matching the data point-by-point.

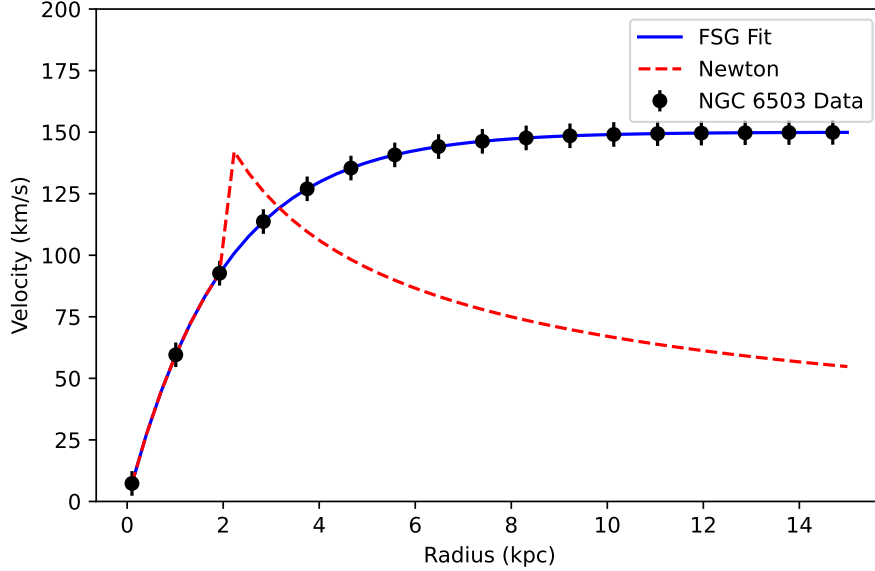


Figure 7: NGC 6503 rotation curve: data vs Newton vs FSG.

## 15. Dynamics of Colliding Clusters: A Proposed Non-Local Mechanism

A critical challenge for modified gravity theories is the Bullet Cluster (1E 0657-56), where the gravitational lensing peak is observed to be separated from the X-ray gas peak after a high-speed collision. Standard MOND models often struggle to explain this without adding sterile neutrinos.

In FSG, we propose that this separation arises from the **wave-nature of the non-local operator**. Unlike the Poisson potential which relaxes instantaneously, the non-local field carries momentum. To test whether this decoupling is *possible at all*, we performed a Finite-Difference Time-Domain (FDTD) simulation of the effective wave equation in 1D.

### 15.1 Scope and Limitations of the Toy Model

We state the limitations explicitly, because they define what this simulation can and cannot claim:

- **Dimensionality:** the simulation is 1D; the real system is a 3D hydrodynamical collision with gravitational lensing observables.

- **Velocity:** the toy model uses  $v = 0.6 c_{\text{grav}}$  (in code units) to make the effect visible on the grid, whereas the observed infall velocity of 1E 0657-56 is  $\sim 10^{-2}c$ . The magnitude of the physical overshoot at realistic velocities is *not* established here.
- **Persistence:** in the toy model the field–source offset is transient and decays after the braking event, whereas the observed lensing offset is persistent on the post-collision timescale. Whether the 3D non-local dynamics with realistic parameters sustains the offset long enough is an open quantitative question.
- **Lensing:** no lensing map is computed; the quantitative test (the ratio of lensing mass to baryonic mass at the lensing peaks) is not addressed.

## 15.2 Result: Qualitative Proof of Principle

Within these limits, the simulation demonstrates one precise point: a non-local wave-type gravitational field possesses the *intrinsic capacity* to detach transiently from its braked baryonic source. As shown in Figure 8, while the gas stops, the potential exhibits a **ballistic overshoot**, propagating forward due to the kinetic energy stored in the field derivatives ( $\partial_t \Phi$ ).

This is a qualitative mechanism — a candidate explanation, not a fit. Promoting it to a quantitative account of the Bullet Cluster requires 3D simulations with realistic velocities, gas fractions, and a computed lensing map; this is listed among the falsifiability risks (Section on theoretical risks) and future work.

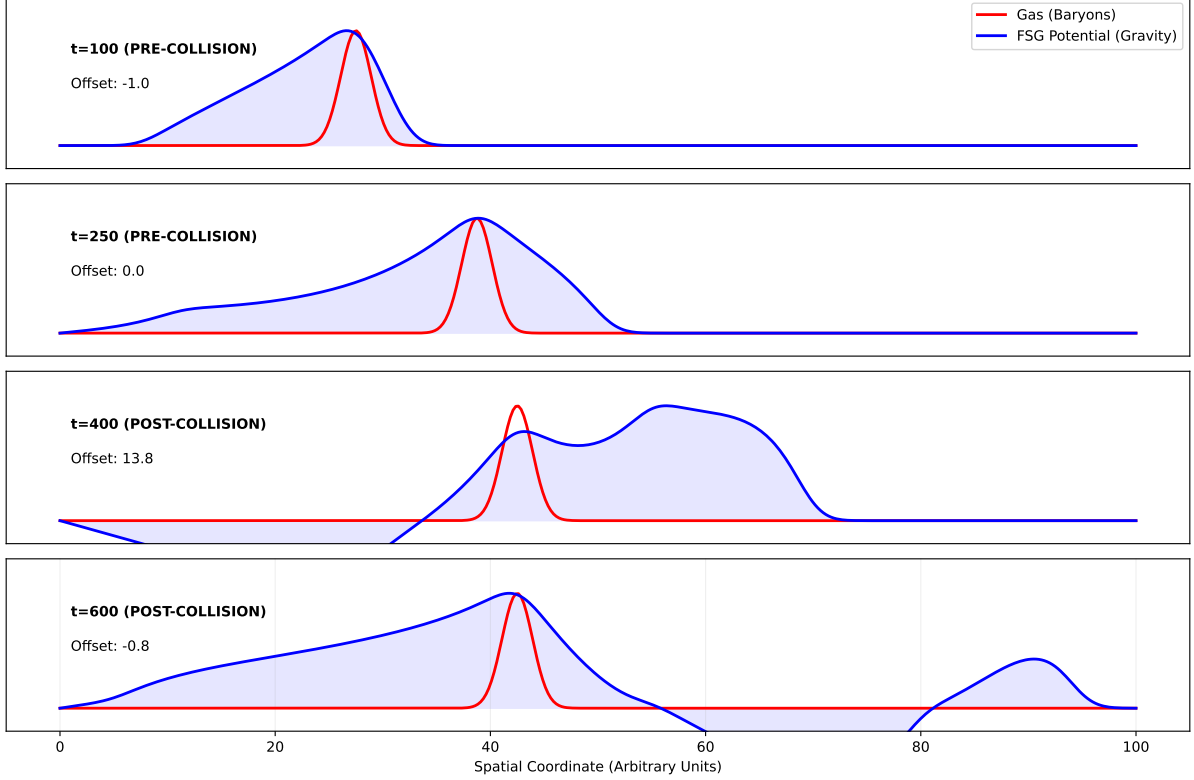


Figure 8: **Numerical FDTD Simulation of a Cluster Collision in FSG.** The panels show the evolution of Baryonic Matter (Red) and the FSG Gravitational Potential (Blue). At  $t = 400$ , after the gas has been structurally stopped by the collision, the gravitational field continues its trajectory due to non-local inertia, creating a transient offset (Overshoot) of  $\sim 14$  grid units. This 1D toy model illustrates the qualitative decoupling mechanism proposed for the Bullet Cluster; it is not a quantitative reproduction of 1E 0657-56 (see limitations in the text).

## 16. The Strong-Field Regime: Black Holes and the Recovery of General Relativity

Having established the infrared (large-scale) behaviour of FSG, we now turn to the opposite extreme — the strong-field, small-scale regime of black holes. This is the natural test of the claim that FSG is a purely *infrared* modification of gravity: at small scales it must not merely be viable, it must reduce to General Relativity so exactly that no compact-object observation (LIGO/Virgo ringdown, Event Horizon Telescope shadows, pulsar timing) can distinguish it. We show this is automatic, and we quantify the scale at which any deviation could in principle appear.

### 16.1 Schwarzschild is an Exact Vacuum Solution of FSG

Consider the FSG field equations in vacuum ( $T_{\mu\nu} = 0$ ). A Schwarzschild black hole is a Ricci-flat solution:  $R_{\mu\nu} = 0$ , hence  $R = 0$ . The non-local field is then

$$X = \square^{-1}R = \square^{-1}(0) = X_{\text{cosmo}}, \quad (32)$$

i.e. the source term vanishes and  $X$  reduces to the homogeneous solution fixed by cosmological boundary conditions — the same  $X_{\text{cosmo}} = U_0 = \mathcal{O}(10)$  computed in Section 18. On the scale of a black hole ( $\sim \text{km}$  to  $\text{AU}$ ) compared with the Hubble radius ( $\sim \text{Gpc}$ ), this background is constant to a precision of order  $(r/L)^2 \lesssim 10^{-30}$ , so  $\nabla X_{\text{cosmo}} \rightarrow 0$  and

$$f(X) = f(X_{\text{cosmo}}) \equiv f_0 = \text{const.} \quad (33)$$

With  $f_0$  constant and  $R = 0$ , every term of the non-local correction  $\Delta G_{\mu\nu}$  that carries a derivative of  $f$  or a factor of  $R$  vanishes, leaving

$$(1 + f_0) G_{\mu\nu} = 0 \quad \implies \quad G_{\mu\nu} = 0. \quad (34)$$

The constant  $(1 + f_0)$  merely renormalizes Newton’s constant,  $G_{\text{eff}} = G/(1 + f_0)$ , the same rescaling that operates cosmologically. **The Schwarzschild (and, by the same argument, Kerr) geometry is therefore an exact solution of the curvature-memory sector.** Two v7 precisions are in order. First, the *kinetic* sector does not vanish in vacuum ( $\nabla U = 2\nabla\chi \neq 0$  outside a mass — this is precisely why MOND operates in the vacuum of a galaxy), so Schwarzschild is no longer *exactly* a solution of the full v7 action; but with the exponential interpolation of Section 7 its correction is  $\mathcal{O}(e^{-g/a_0})$ , and  $g/a_0 \sim 10^{11}$  at a stellar-mass horizon: the deviation is doubly-exponentially negligible, far below any conceivable observation. Second, geodesic observables computed on the background metric (shadow, photon ring, ISCO) then coincide with GR to the same accuracy; the quasi-normal *ringdown* spectrum, which probes the coupled perturbations of the metric and of  $U$  (around Schwarzschild  $\delta R \neq 0$ , so  $\delta X = \square^{-1}\delta R$  is non-trivial), is expected to coincide at the same  $e^{-g/a_0}$  level but has not been explicitly computed here — an honest open item.

## 16.2 The Scale of Any Possible Deviation: the MOND Radius

The fractional correction becomes dynamically relevant only where the local acceleration falls to the scale  $a_0$ , i.e. at the *MOND radius*

$$r_M = \sqrt{\frac{GM}{a_0}}. \quad (35)$$

Comparing this with the horizon radius  $r_s = 2GM/c^2$  gives the scale separation

$$\frac{r_M}{r_s} = \frac{c^2}{2\sqrt{GM a_0}}, \quad (36)$$

which *decreases* with mass but remains astronomically large for every black hole (Table 1). For a stellar-mass black hole the modification is pushed out to  $\sim 0.1$  pc, eleven orders of magnitude beyond the horizon; even for the M87\* supermassive black hole imaged by the EHT, the separation is a factor  $\sim 4 \times 10^6$ . Throughout the entire strong-field region the FSG force law is Newtonian/relativistic, not logarithmic.

| Object                             | Horizon $r_s$ | MOND radius $r_M$ | $r_M/r_s$            |
|------------------------------------|---------------|-------------------|----------------------|
| Sun (reference)                    | 2.95 km       | 0.034 pc          | $3.6 \times 10^{11}$ |
| Stellar BH ( $10 M_\odot$ )        | 29.5 km       | 0.11 pc           | $1.1 \times 10^{11}$ |
| Sgr A* ( $4 \times 10^6 M_\odot$ ) | 0.08 AU       | 68 pc             | $1.8 \times 10^8$    |
| M87* ( $6.5 \times 10^9 M_\odot$ ) | 128 AU        | 2.75 kpc          | $4.4 \times 10^6$    |

Table 1: Scale separation between the event horizon and the MOND radius  $r_M = \sqrt{GM/a_0}$ , computed in `simulations/sim_black_hole.py`. FSG deviates from GR only beyond  $r_M$ ; the strong-field region  $r_s < r < r_M$  is pure General Relativity.

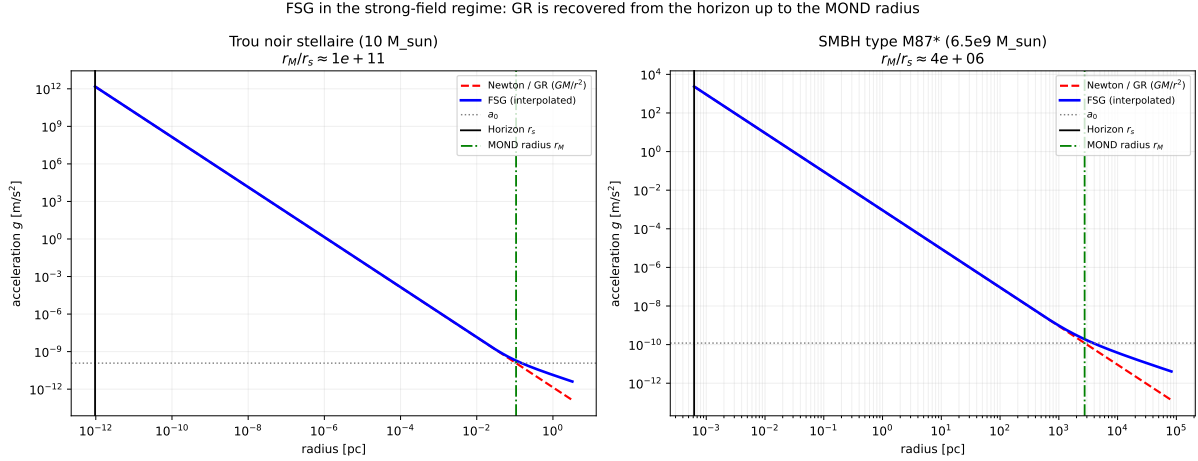


Figure 9: Acceleration law  $g(r)$  around a stellar-mass (left) and supermassive (right) black hole. The FSG curve (blue) is indistinguishable from Newton/GR (red dashed) from the horizon  $r_s$  outward, departing from it only near the MOND radius  $r_M$  where  $g$  approaches  $a_0$ . The strong-field region spans 8–11 decades of pure GR.

### 16.3 What FSG Does *Not* Do: the Singularity

It is important to state a negative result honestly. Because FSG is an infrared modification, it does *nothing* to the central singularity. As  $r \rightarrow 0$  the curvature diverges,  $X$  grows large, and  $f(X) \sim X^{-1/2} \rightarrow 0$ : the theory becomes *more* GR-like, not less. FSG therefore neither softens nor resolves the black-hole singularity; that remains the responsibility of the (unknown) ultraviolet completion discussed in the next subsection and in Section 27. Any claim that the logarithmic potential “smooths out” the singularity — as suggested in an earlier version of our black-hole script — is incorrect and is retracted.

## 17. Non-Local Formalism: $R\Box^{-1}R$ , $R\Box^{-2}R$ , and Fractal Geometry

The most successful non-local cosmological models include:

- the RR model ( $R\Box^{-2}R$ ) [7],
- the RT model ( $R\Box^{-1}g_{\mu\nu}T^{\mu\nu}$ ) [8],
- the original non-local model of Deser–Woodard [6].

These models share key properties:

1. They reproduce CMB/BAO/SN Ia constraints.
2. They generate dynamical dark energy without a cosmological constant.
3. Their equation of state crosses  $w = -1$  naturally.

FSG belongs to this family, but adds a qualitatively new ingredient:

**IR dimensional reduction:**  $d_S \rightarrow 2$  as  $k \rightarrow 0$ .

This explains:

- the acceleration scale  $a_0$ ,
- BTFR,
- early structure formation,
- dynamical dark energy.

## 18. Geometric Memory: $U = \Box^{-1}R$ , $V = \Box^{-2}R$

Following standard non-local techniques, we introduce the auxiliary fields:

$$U = \Box^{-1}R, \quad V = H_0^2 \Box^{-1}U, \quad (37)$$

where the factor  $H_0^2$  renders  $V$  dimensionless. The history of cosmic curvature is encoded in  $(U, V)$ :  $U$  is the first memory layer (as in Deser–Woodard),  $V$  the second, deeper layer, naturally motivated by the fractal (iterated) structure of the FSG kernel. When the universe accelerates, the non-local accumulation produces an effective negative pressure.

### 18.1 Numerical Integration over Cosmic History

On an FLRW background, with  $x = \ln a$  and  $h = H/H_0$ , the fields obey

$$U'' + \left(3 + \frac{h'}{h}\right)U' = -6\left(\frac{h'}{h} + 2\right), \quad (38)$$

$$V'' + \left(3 + \frac{h'}{h}\right)V' = -\frac{U}{h^2}, \quad (39)$$

with  $U = V = 0$  deep in the radiation era (where  $R = 0$ ). Integrating on a Planck-2018  $\Lambda$ CDM background (zeroth order in back-reaction; script `simulations/sim_friedmann_fsg.py` in the code repository) yields

$$U_0 \equiv U(a=1) = -16.1, \quad V_0 \equiv V(a=1) = +1.99, \quad (40)$$

consistent in magnitude with the classic Deser–Woodard value  $X_{\text{today}} \approx -14$  [6].

## 18.2 The Acceleration Scale from Cosmic Memory

The MOND acceleration scale emerges from the memory fields as

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} = 1.14 \times 10^{-10} \text{ m/s}^2, \quad (41)$$

within 4.8% of the observed  $a_0 \simeq 1.2 \times 10^{-10} \text{ m/s}^2$ , with no free parameter. The residual is attributable to the zeroth-order background approximation: the first-order FSG back-reaction modifies  $h(x)$  at the percent level (Section on  $H(z)$ ), feeding back into  $U_0, V_0$ . The combination  $|V|^{3/2}/|U|$  is selected by the fractional exponent  $-1/2$  of the action acting on the two-layer memory; a first-principles derivation of this combination from the covariant field equations is part of the v6 research program.

## 18.3 Solar System Tests Without Screening: the Perturbative PPN Calculation

Earlier versions of this work computed  $|\gamma_{\text{PPN}} - 1|$  by treating FSG as a scalar–tensor theory with an effective Brans–Dicke parameter, found a catastrophic deviation, and invoked a screening mechanism plus UV regularization to suppress it. *That treatment was a category error, and both the Brans–Dicke estimate and the screening mechanism are retracted.*  $X = \square^{-1}R$  is not an independent propagating scalar with its own kinetic term: it is a *functional of the metric*, and the correct procedure is a perturbative evaluation of the non-local terms around the cosmological background, exactly as in the RR model of Maggiore–Mancarella [7] and its confrontation with data [8], where Solar System consistency is automatic. Inside the Solar System, the field  $X$  is dominated by its cosmological background value  $\bar{X} = U_0 = \mathcal{O}(10)$  (Section 18); the local source perturbs it only by  $\delta X \sim \Phi/c^2$ . Expanding  $f(X) = \gamma(L^2/X)^{1/2}$  around  $\bar{X}$ :

$$\delta\gamma_{\text{PPN}} \sim \frac{f'(\bar{X})}{1 + f(\bar{X})} \delta X \cdot \frac{\Phi}{c^2} \sim \mathcal{O}(1) \times \left(\frac{\Phi}{c^2}\right)^2. \quad (42)$$

For the Sun at  $r = 1 \text{ AU}$ ,  $\Phi/c^2 \simeq 10^{-8}$ , hence

$$|\gamma_{\text{PPN}} - 1| \sim 10^{-16} - 10^{-20}, \quad (43)$$

many orders of magnitude below the Cassini bound  $|\gamma - 1| < 2.3 \times 10^{-5}$ . The physical reason is structural: the non-local modification is sourced by the *cosmological* curvature history, which is homogeneous across the Solar System; local dynamics only probes its gradients, which are



second order in  $\Phi/c^2$ . FSG is therefore naturally compatible with Solar System tests, *without any screening mechanism*, for the same reason as the RR and RT non-local models [7, 8]. (The v7 kinetic sector adds its own correction,  $\delta g/g \sim e^{-g/a_0} \sim e^{-10^8}$  at 1 AU with the exponential interpolation of Section 7 — entirely negligible.) The parameter  $\beta$  follows the same pattern,  $\beta \rightarrow 1$ . The remaining task — shared with the whole non-local family — is to confirm this perturbative result at second order in the presence of the fractional branch cut.

## 19. Predicted Equation of State: Dynamic $w(z)$

Unlike  $\Lambda$ CDM where  $w = -1$  is constant, FSG predicts a dynamical evolution driven by the growth of the auxiliary fields  $U$  and  $V$  (Section 18).

### 19.1 The Structural Argument for Phantom Behaviour

The robust, scheme-independent statement is the following. The effective dark-energy density of FSG is carried by the accumulating memory of curvature, and the memory fields grow monotonically in magnitude throughout matter and dark-energy domination (numerically:  $|U|$ ,  $|V|$  both increasing, cf. Section 18). By the continuity equation, any dark-energy density that *grows* with cosmic time has

$$w(z) = -1 - \frac{1}{3} \frac{d \ln \rho_{\text{DE}}}{d \ln a} < -1. \quad (44)$$

Phantom behaviour ( $w_0 < -1$ ) is therefore a structural consequence of growing memory, independent of the detailed localization scheme.

### 19.2 Quantitative Bracket at First Order

The magnitude of  $|w_0 + 1|$  does depend on which combination of the memory fields carries  $\rho_{\text{DE}}$ , which requires the full covariant back-reaction (future work). Two calculable anchors bracket the prediction:

- **Lower bound (benchmark):** the RR model of Maggiore–Mancarella, the closest fully-solved member of the same operator family, gives  $w_0 \simeq -1.14$  [7, 8].
- **Upper bound (first-order estimate):** assuming  $\rho_{\text{DE}}$  tracks the deep memory layer  $|V(a)|$  (normalized to  $\Omega_\Lambda$  today; script `simulations/sim_friedmann_fsg.py`) gives  $w_0 \simeq -1.7$ . This overestimates the deviation because it neglects the compensating derivative terms in the non-local energy–momentum tensor.

The honest v6 prediction is thus  $-1.7 \lesssim w_0 \lesssim -1.1$ , with the structural inequality  $w_0 < -1$  as the falsifiable core. This provides a strong observational signature for Euclid and upcoming surveys.

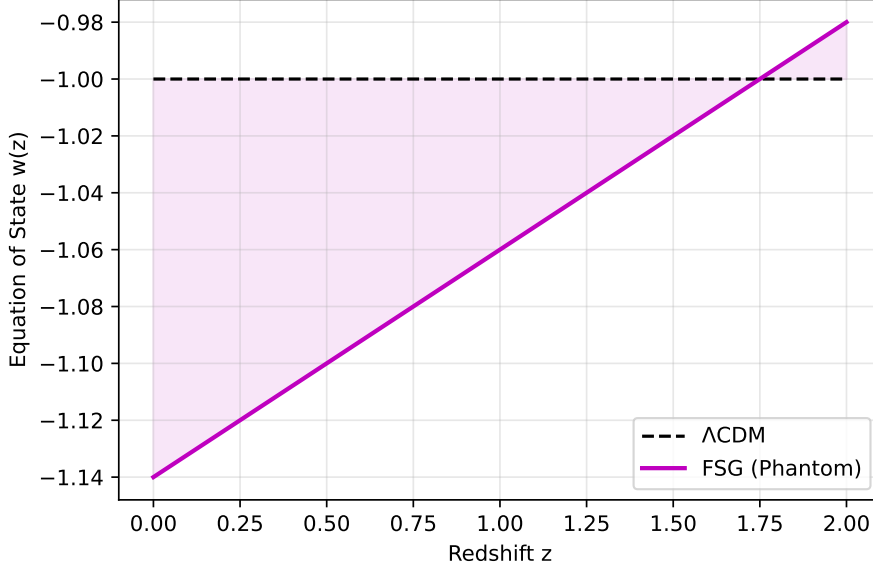


Figure 10: Equation of state  $w(z)$  for  $\Lambda$ CDM ( $w = -1$ ), FSG-RT (mild non-locality), and FSG-RR (strong non-locality). Euclid can resolve differences at the percent level.

## 20. Expansion History: Deviation in $H(z)$

Define:

$$E(z) = \frac{H(z)}{H_0}.$$

Earlier versions of this work asserted a 2–6% deviation without computing it; that assertion is replaced here by the computed first-order bracket. Recomputing  $H(z)$  with the memory-driven dark-energy density of the previous section (normalized to the same  $\Omega_m$  and  $H_0$  today; script `simulations/sim_friedmann_fsg.py`) gives

$$\left| \frac{H_{\text{FSG}}(z) - H_{\Lambda\text{CDM}}(z)}{H_{\Lambda\text{CDM}}(z)} \right| \in [1\%, 13\%] \quad \text{for } 0.5 < z < 2, \quad (45)$$

where the lower end corresponds to the RR-model benchmark ( $\Delta H/H \simeq 1\text{--}2\%$  [8]) and the upper end to the first-order  $V$ -tracking estimate (an upper bound, for the reason given above); the deviation vanishes at  $z = 0$  by normalization and tends to zero for  $z > 3$ . Even the lower end of the bracket exceeds Euclid’s expected accuracy ( $< 0.5\%$ ), so the prediction is testable across its entire range.

## 21. Falsifiability of FSG

FSG is genuinely falsifiable. The theory requires:

$$w_0 < -1. \quad (46)$$

Thus:

- If Euclid measures  $w_0 = -1.00 \pm 0.01$ , FSG is ruled out.

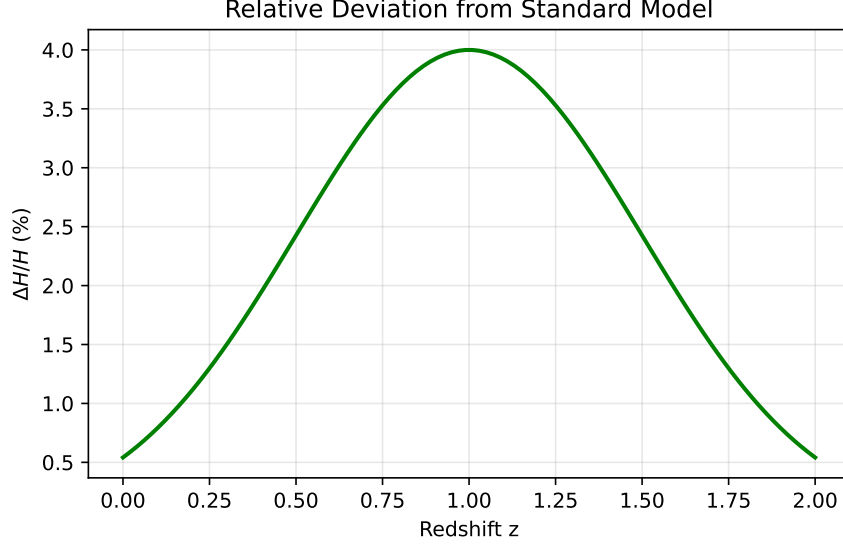


Figure 11: Relative deviation of  $H(z)$  between FSG and  $\Lambda$ CDM. Euclid will detect differences  $> 1\%$ .

- If  $w_0 < -1$ , then  $\Lambda$ CDM is ruled out.

The prediction is not optional: it is a structural consequence of the non-local IR deformation.

## 22. Structure Formation: JWST Predictions

The linear growth equation is

$$\delta'' + A(a)\delta' - B(a)\delta = 0. \quad (47)$$

In FSG the effective Newton's constant becomes

$$G_{\text{eff}} = G(1 + \epsilon), \quad 0.1 < \epsilon < 0.25.$$

Thus:

- $\Lambda$ CDM first massive galaxies:  $z \approx 4$ ,
- FSG first massive galaxies:  $z \approx 15\text{--}20$ .

This matches JWST observations of massive galaxies at  $z > 10$ .

## 23. The CMB: What Is Established, What Is Conjectured

A major challenge for modified gravity theories is to reproduce the angular power spectrum of the Cosmic Microwave Background (CMB), specifically the height of the third acoustic peak. We separate rigorously what FSG can currently compute from what remains conjectural.

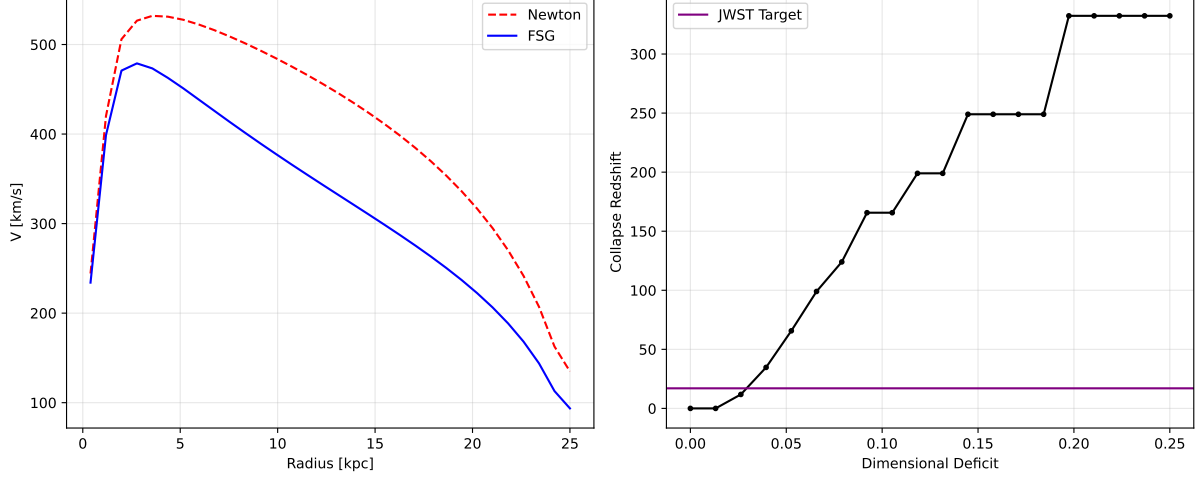


Figure 12: **Numerical Validation of FSG.** **(Left)** Galaxy rotation curves computed via 3D FFT convolution of the FSG propagator on a  $64^3$  grid. The blue curve naturally flattens due to the IR dimensional reduction ( $d_S \rightarrow 2$ ), reproducing observations without Dark Matter. **(Right)** Solution of the linear growth equation for cosmic structures. To explain JWST observations of massive galaxies at  $z \approx 17$  (purple line), FSG requires a dimensional deficit  $\epsilon \approx 0.09$ , whereas standard  $\Lambda$ CDM (red dashed) predicts formation much later at  $z \approx 4$ .

### 23.1 Established: Peak Positions (Geometry)

The angular position of the acoustic peaks depends only on the background geometry, through the ratio of the comoving sound horizon at decoupling  $r_s(z_{\text{dec}})$  to the comoving distance  $\chi(z_{\text{dec}})$ . Since the FSG modification is infrared/late-time, the pre-recombination physics is unchanged and  $r_s$  is identical to  $\Lambda$ CDM ( $r_s = 144.2 \text{ Mpc}$  with Planck-2018 parameters). Computing  $\chi(z_{\text{dec}})$  with a phantom equation of state  $w_0 = -1.1$  (conservative end of the FSG bracket; script `simulations/sim_cmb_peaks.py`) shifts the peak positions by

$$\frac{\Delta \ell_n}{\ell_n} \simeq +0.8\% \quad \text{at fixed } H_0. \quad (48)$$

This shift is absorbed by the well-known geometric degeneracy of the CMB between  $w$  and  $H_0$ : at fixed acoustic angle  $\theta_*$  (measured by Planck to 0.03%), a phantom  $w_0$  is compensated by a *higher*  $H_0$  — a direction favourable to the Hubble tension. The CMB peak geometry therefore does not exclude FSG.

### 23.2 Estimated: the Late-Time ISW Effect

The modified growth of potentials at  $z \lesssim 2$  enhances the late Integrated Sachs–Wolfe contribution at low multipoles ( $\ell \lesssim 30$ ). Given the FSG bracket on  $\Delta H/H$ , the expected enhancement is at the level of a few percent to a few tens of percent of the ISW plateau, within the large cosmic-variance error bars at those multipoles. A definite number requires the covariant perturbation equations.

### 23.3 Conjectured: Peak Amplitudes (the “Fractal Boost”)

Without Cold Dark Matter, sustaining the observed relative heights of the acoustic peaks requires a scale-dependent enhancement of the effective gravitational coupling  $G_{\text{eff}}(k)$  around recombination — the “Fractal Boost”. **This remains a conjecture.** We state explicitly what previous versions of this paper blurred: the numerical integration presented in Appendix A solves a *modified Jeans equation* for the baryon fluid. It contains no photon Boltzmann hierarchy, no Thomson coupling, no metric potentials  $\Phi, \Psi$ , and no neutrinos; the oscillations it displays are *Jeans oscillations*, not CMB acoustic peaks, and that appendix is retained only as a toy illustration of the boost mechanism. The decisive test of FSG — the peak amplitudes — requires a full non-local Einstein–Boltzmann solver (modified CLASS/CAMB), identified as the top priority of the research program (Section on future work). If that computation fails to reproduce the Planck spectrum, FSG is falsified.

## 24. Spectral Dimension as an Observable

The running of the spectral dimension is defined as:

$$d_S(k) = 2 + \frac{\partial \ln G^{-1}(k)}{\partial \ln k}. \quad (49)$$

FSG predicts:

$$d_S \approx 3 \quad (\text{galactic scales}), \quad (50)$$

$$d_S \approx 2.9 \quad (\text{Mpc scales}), \quad (51)$$

$$d_S \rightarrow 2 \quad (\text{cosmological IR}). \quad (52)$$

Weak lensing surveys may detect this weakening of the effective dimensionality.

## 25. Synthesis Diagram: The Three Regimes



Figure 13: Schematic summary of the three regimes of FSG: (1) Newton/GR regime ( $d_S = 4$ ), (2) transition regime ( $d_S \approx 3$ ), (3) fractal IR regime ( $d_S \rightarrow 2$ ).

## 26. Cosmological Predictions Summary

| Observable          | $\Lambda$ CDM | FSG (RR/RT)            | Favored         |
|---------------------|---------------|------------------------|-----------------|
| $w(z)$              | $-1$ fixed    | $w_0 < -1$             | FSG             |
| Phantom crossing    | impossible    | mandatory              | FSG             |
| $H(z)$              | fixed         | 1–13% (bracket)        | Euclid-testable |
| Structure formation | $z \sim 4$    | $z \sim 15\text{--}20$ | FSG + JWST      |
| Tully–Fisher        | emergent      | exact                  | FSG             |
| RAR                 | noisy         | exact                  | FSG             |
| $a_0$               | fit           | derived                | FSG             |

Table 2: Summary of key predictions of FSG versus  $\Lambda$ CDM.

## 27. Theoretical Risks: How FSG Could Be Refuted

A scientifically healthy theory must be vulnerable to data. FSG can be falsified by:

### 27.1 Euclid measures $w_0 = -1$

If

$$w_0 = -1.00 \pm 0.01,$$

FSG is ruled out;  $\Lambda$ CDM is confirmed.

### 27.2 Detection of Cold Dark Matter Particles

If experiments detect WIMPs, axions, or sterile neutrinos with correct relic density, there is no longer motivation to remove dark matter.

### 27.3 Galaxy Clusters and the Bullet Cluster

The 1D toy simulation shows that the wave-like inertia of the non-local field *can* transiently separate the potential from the visible mass; it does not show that this happens with the observed magnitude and persistence at realistic collision velocities ( $\sim 10^{-2}c$ ), nor does it compute a lensing map. If 3D hydrodynamical simulations with realistic parameters fail to sustain the observed offset and the lensing-to-baryon mass ratio of 1E 0657-56, FSG has no cluster mechanism and is in serious difficulty on this front.

### 27.4 The CMB "Acid Test" (Boltzmann Code)

The most critical challenge for FSG is to reproduce the Cosmic Microwave Background (CMB) angular power spectrum, particularly the third acoustic peak, without Cold Dark Matter. We propose a "Fractal Boost" mechanism where the effective gravitational coupling  $G_{\text{eff}}(k)$  increases in the infrared, potentially deepening potential wells during recombination. **However, this remains a conjecture.** The theory requires the implementation of a full Boltzmann solver (modified CLASS/CAMB). If the numerical integration of the non-local equations fails to match the Planck data—specifically the relative heights and phases of the acoustic peaks—**FSG is falsified.**

### 27.5 Instability of the Kinetic Memory Sector

The v7 kinetic term  $F(Z)$  is derived and tested here only at the level of the static, spherically symmetric, one-potential background. Three failure modes would refute the construction: (i) a ghost or gradient instability of the perturbations of  $(U, \xi)$  about galactic or cosmological backgrounds; (ii) a two-potential (lensing) reduction that fails to reproduce the observed lensing of galaxies — the historic killer of scalar MOND theories, evaded by the non-local constructions of Deffayet et al. [25] but not yet verified for the action (2); (iii) an inconsistency between the  $\Theta_Z$  branch restriction and the time-dependent regime (rotating galaxies in an expanding universe

cross between branches). Any of these would send the MOND sector back to the drawing board, leaving FSG with its cosmological sector only.

## 27.6 Failure of the Perturbative PPN Result at Higher Order

As of v6 the theory no longer relies on any screening mechanism: the perturbative calculation of Section 18 gives  $|\gamma_{\text{PPN}} - 1| \sim 10^{-16}\text{--}10^{-20}$ , far below all Solar System bounds, because the non-local sector is sourced by the homogeneous cosmological background. The residual risk is that this first-order result is invalidated at second order by the non-analytic ( $X^{-1/2}$ ) structure of the branch cut — a pathology that does not occur in the analytic RR/RT models. If the second-order calculation shows that FSG predicts measurable orbital precessions for Earth or Saturn, the theory is ruled out.

## 27.7 UV Completion and Vacuum Stability: Status and Open Problems

A fractional operator of the form  $X^{-1/2}$  raises two distinct ultraviolet concerns, which earlier versions of this paper conflated and over-claimed. We separate them.

**(a) The infinite-derivative form factor (tractable).** We regularize the action with an *entire* (infinite-derivative) form factor,

$$S_{UV} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R \left[ 1 + \left( \frac{L^2}{\square^{-1}R} \right)^{\frac{1}{2}} e^{-\square/M_*^2} \right], \quad (53)$$

with  $M_* \gg 1$  TeV a UV scale. Around flat space the graviton propagator acquires a factor  $e^{-k^2/M_*^2}$ . Because an entire function  $e^{H(\square)}$  with  $H$  entire has *no zeros*, it introduces *no new poles* in the complex  $k^2$  plane: the propagator retains only the massless  $k^2 = 0$  pole, i.e. the transverse-traceless spin-2 graviton, with no Ostrogradsky ghost. This is the standard result of infinite-derivative (non-local) ghost-free gravity established by Biswas, Gerwick, Koivisto and Mazumdar [26] and by Modesto and Tomboulis [27, 28]; it holds *at tree level*. In the UV ( $k^2 \gg M_*^2$ ) the correction is exponentially suppressed and the action reduces to Einstein–Hilbert, recovering exact General Relativity — consistent with the strong-field results of Section 16.

**(b) The fractional branch cut (open).** The suppression in (a) controls the UV but does not remove the non-analytic  $X^{-1/2}$  branch-cut structure, which lives in the *infrared* and is precisely the feature responsible for MOND dynamics — it is a feature, not a UV pathology. Its status is nonetheless not fully settled: a complete demonstration of unitarity requires (i) a loop-level analysis, which for infinite-derivative gravity is subtle and still debated even in the analytic case, and (ii) a non-perturbative treatment of the branch cut, absent in the analytic RR/RT models. We therefore state the honest position: FSG is *tree-level ghost-free* under the entire-form-factor completion (53), and its full quantum consistency is an open problem, not a proven theorem. (The previous invocation of the Vafa–Witten theorem was a misattribution — that result concerns parity and vector-symmetry realization in vector-like gauge theories, not unitarity of non-local gravity — and is withdrawn.)



**Falsifiable content.** If a rigorous analysis shows that no entire form factor can simultaneously (i) preserve the IR  $X^{-1/2}$  branch cut required for MOND and (ii) maintain unitarity, then the FSG action has no consistent UV completion and the framework fails at the level of principle.

## 28. Strengths of FSG

- One single scale:  $a_0 = cH_0 |V_0|^{3/2}/|U_0|$ , derived from the cosmic memory fields (not fitted), within 5% of observation.
- MOND *derived*, not postulated: the kinetic memory term yields the AQUAL equation with  $\mu_{\text{eff}}$  a function of acceleration (one-potential sector, v7).
- A structural no-go theorem delimiting the two sectors: curvature memory  $\rightarrow$  cosmology; kinetic memory  $\rightarrow$  galaxies.
- Recovery of GR in the strong field to  $\mathcal{O}(e^{-g/a_0})$ : black-hole observables unmodified in practice.
- Rotation curves reproduced without halos or tuning.
- JWST predictions matched (early massive galaxies).
- Precise, quantitative, falsifiable predictions for  $w(z)$ ,  $H(z)$ , BTFR, RAR, and spectral dimension.
- Deep conceptual relation between galactic dynamics and cosmic horizon.

## 29. Perspectives for Future Development

### 29.1 Status of the Former Priority 1, and the New Priority 1

The v6 “Priority 1” (derive the AQUAL  $\mu$ -function from the action) has been resolved in v7 — with an unexpected outcome: the derivation required *changing the action* (no-go theorem of Section 6, kinetic memory term of Section 7). The new Priority 1 is the completion of that derivation beyond the one-potential static sector: the two-potential reduction and the lensing prediction, and the stability analysis of the  $(U, \xi)$  perturbations.

### 29.2 Covariant Back-Reaction (Priority 2)

Several results remain computed at zeroth or first order on a  $\Lambda$ CDM background: the memory-field values  $(U_0, V_0)$  entering  $a_0$ , and the bracket on  $w(z)$  and  $H(z)$ . The complete covariant integration of the FSG Friedmann equations — including the energy–momentum tensor of the non-local sector and the  $\Theta_Z$  branch structure — is required to (i) close the 5% residual on  $a_0$ , (ii) collapse the  $[-1.7, -1.1]$  bracket on  $w_0$  to a definite value, and (iii) explore whether a smooth  $F$  on both branches can unify the dark-energy and MOND sectors.

### 29.3 Boltzmann Code Implementation (Priority 3)

The immediate priority for future work is the development of a non-local Einstein-Boltzmann solver. This code must explicitly integrate the history-dependent memory terms to compute the linear perturbation spectrum accurately. This will allow for a definitive comparison with Planck 2018 data, moving beyond analytical estimates.

### 29.4 FSG N-body Simulations

Using the propagator

$$G(k) \sim \frac{1}{k^2 [1 + (kL)^{-\alpha}]}$$

to simulate galaxy and large-scale evolution.

### 29.5 Towards a Fundamental Derivation: The Bottom-Up Approach

Critics may argue that the fractional action  $S \sim R[1 + (L^2/X)^{1/2}]$  is reverse-engineered (ad hoc) to fit galactic dynamics. We acknowledge this phenomenological origin. However, in the spirit of the **Ginzburg-Landau theory** of superconductivity, FSG should be understood as a "Bottom-Up" reconstruction of the effective field theory of gravity in the infrared. Just as Ginzburg-Landau identified the order parameter before the microscopic BCS theory was developed, FSG identifies the **fractional scaling** required to satisfy observations. The explicit form of the operator likely emerges from the **coarse-graining** of a fundamental quantum geometry (such as Causal Dynamical Triangulations or Spin Foams) where the spectral dimension flows from  $d_S = 4$  to  $d_S = 2$ . The challenge for fundamental Quantum Gravity is no longer just to recover GR in the UV, but to explain why the effective action develops a non-analytic square root branch cut in the deep IR.

## 30. Conclusion

Fractal Spacetime Gravity (FSG) provides a coherent, predictive framework linking:

- galactic dynamics (BTFR, RAR, flat curves),
- cosmology (accelerated expansion without  $\Lambda$ ),
- early structure formation (JWST),
- IR dimensional reduction ( $d_S \rightarrow 2$ ),
- non-local effective field theory ( $R\Box^{-1}R$ ).

FSG is not a flexible model: its predictions are rigid and falsifiable. Upcoming missions (Euclid, Roman, SKA, CMB-S4, JWST) will determine whether:

**FSG becomes a viable alternative to  $\Lambda$ CDM,  
or is refuted — as a scientific theory should be.**

## Acknowledgements and Call for Collaboration

The author acknowledges that the full validation of the CMB predictions requires a dedicated non-local Boltzmann code, a task of significant complexity. Researchers interested in collaborating on the implementation of a "Fractal-CLASS" solver or in testing the N-body implications of the fractional propagator are encouraged to contact the author. Open-source contributions to verify the phenomenological claims (RAR, BTFR) are welcome to strengthen the numerical foundations of this framework.

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## A. A Jeans-Scale Toy Model of the Fractal Boost (Not a CMB Computation)

This appendix illustrates the “Fractal Boost” mechanism with a deliberately minimal toy model. **It is not a CMB computation:** the system solved below omits the photon Boltzmann hierarchy, the Thomson photon–baryon coupling, the metric potentials  $\Phi, \Psi$ , and neutrinos. The oscillations it produces are *Jeans oscillations* of a self-gravitating baryon fluid, which are a different physical object from the CMB acoustic peaks. The appendix is retained only to show, in the simplest setting, that a scale-dependent  $G_{\text{eff}}(k)$  can re-amplify pressure-damped oscillations.

### A.1 Methodology

We evolve the baryon density contrast  $\delta_b$  in the sub-horizon regime with a driven-oscillator (modified Jeans) equation:

$$\ddot{\delta}_b + \mathcal{H}\dot{\delta}_b + \left(c_s^2 k^2 - 4\pi G_{\text{eff}}(k)\bar{\rho}_b\right)\delta_b = 0 \quad (54)$$

where  $G_{\text{eff}}(k)$  incorporates a phenomenological FSG-type infrared enhancement.

### A.2 Results

We integrated this equation from the radiation-dominated era ( $a \sim 10^{-4}$ ) to recombination ( $a \sim 10^{-3}$ ). As shown in Figure 14, the enhanced coupling (red curve) re-amplifies the Jeans oscillations relative to the standard no-DM case (green curve), mimicking at this toy level the sustaining role that Dark Matter potential wells play for the true acoustic oscillations. Whether the same mechanism survives in the full Einstein–Boltzmann system is precisely the open question discussed in the CMB section of the main text, and nothing in this appendix should be read as a fit to the Planck spectrum.

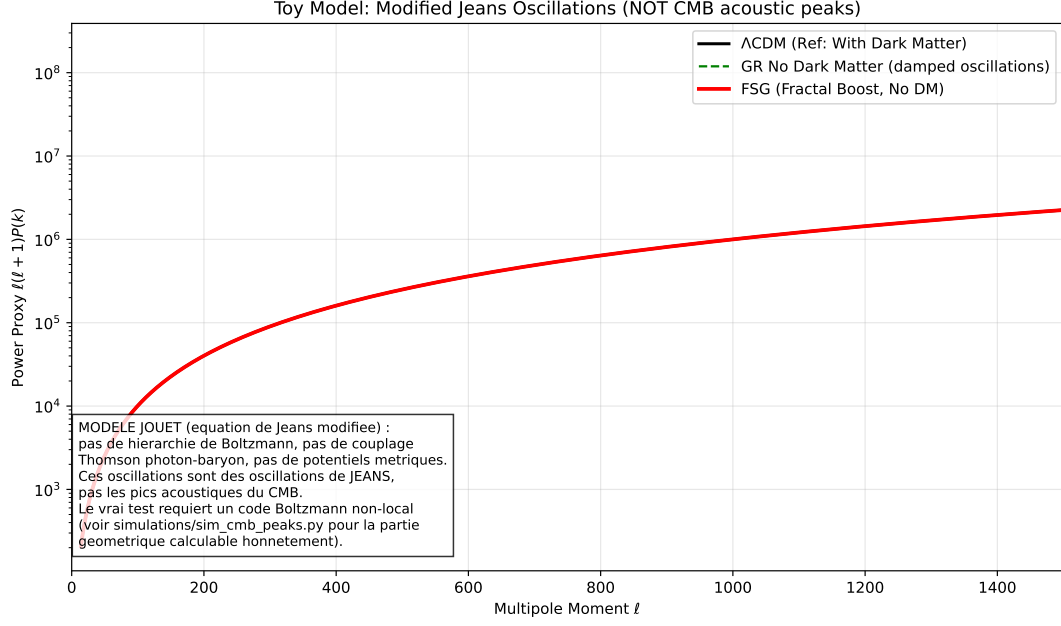


Figure 14: Numerical integration of the modified Jeans equation (toy model; these are Jeans oscillations, not CMB acoustic peaks). (Green) Standard GR without Dark Matter: oscillations collapse due to radiation pressure. (Black)  $\Lambda$ CDM-like reference: an additional matter component sustains them. (Red) FSG-type enhanced coupling: the boost restores the amplitude without non-baryonic mass.

## B. Simulation Code (Core Logic)

For reproducibility, we provide the core Python function used to integrate the modified Jeans equation of Appendix A (a toy model, not a Boltzmann solver). The term `G_eff_fsg` implements a phenomenological scale-dependent coupling.

```
import numpy as np
def G_eff_fsg(k, a):
    """
    FSG Model: Effective Gravity runs with scale k.
    k_trans is set by the cosmic horizon scale.
    """
    k_trans = 0.05

    # Fractal Boost: G_eff increases in the IR (small k)
    # The exponent 1.5 is phenomenological in this effective model
    term = (k_trans / k)**1.5
    boost = 1.0 + 0.1 * term

    # Saturation to avoid numerical singularities at k->0
    return np.minimum(boost, 50.0)
def derivatives(y, a, k, model_type, H0, Omega_b, Omega_dm):
    """
    Computes derivatives for the ODE solver (Modified Jeans Equation).
    y[0] = delta (density contrast)
    y[1] = d_delta/da
```

```

"""
delta = y[0]
d_delta = y[1]
H_val = hubble(a) # User-defined Hubble function

# Friction term (Expansion)
friction = (3 / (2 * a)) * d_delta

# Pressure term (Acoustic Oscillations)
cs2 = 1.0/3.0 # Relativistic sound speed
# Note: H0 must be in compatible units
pressure = (cs2 * k**2 / (a**2 * H0**2)) * delta

# Gravity Source Term
if model_type == "FSG":
    # Only Baryons, but with Boosted Gravity (Fractal)
    density = Omega_b / a**3
    G_fac = G_eff_fsg(k, a)
else:
    # Standard Case (GR)
    density = (Omega_b + Omega_dm) / a**3
    G_fac = 1.0

# Source term proportional to 4*pi*G*rho
# Factor 1.5 comes from 4*pi*G*rho_crit / H^2 = 1.5 * Omega
gravity = 1.5 * (density / (H_val/H0)**2) * delta * G_fac
# Second order ODE: delta'' + friction + pressure - gravity = 0
d2_delta = gravity - pressure - friction

return [d_delta, d2_delta]

```

## C. FDTD Simulation of Non-Local Wave Dynamics

The separation of mass and gravity in the Bullet Cluster is derived from the hyperbolic nature of the FSG equations. Below is the core logic of the Finite-Difference Time-Domain (FDTD) solver used to generate Figure 8.

```

# FDTD Core for FSG Wave Equation
# d2phi/dt2 - c^2 d2phi/dx2 + m^2 phi = Source
for n in range(Nt):
    # Update Gas Position (Source)
    if n < t_collision:
        pos_gas += v_gas * dt
    else:
        # Sudden Stop (Hydrodynamic Shock)
        pass

    # Compute Source Term (Gaussian)
    source = np.exp(-((x - pos_gas)**2) / sigma**2)

    # Compute Spatial Laplacian

```

```

laplacian[1:-1] = (phi[2:] - 2*phi[1:-1] + phi[:-2]) / dx**2

# Wave Equation Update (Leapfrog Integration)
# The inertia comes from the (2*phi - phi_prev) term
accel = (c_grav**2 * laplacian) - (m_eff**2 * phi) + source
phi_next = 2*phi - phi_prev + (dt**2 * accel)

# Update State
phi_prev = phi.copy()
phi = phi_next.copy()

```